# BUSS386 Problem Set 9

Binomial Trees

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## Problem 1 — One step: build the call, then price it

A KOSPI-listed stock trades at ₩ 60,000. In three months it will be either ₩ 69,000 or ₩ 54,000. The 3-month rate is $r = 3\%$. A 3-month European call has $K =$ ₩ 63,000.

(a) Write the call's payoff at each end node.

(b) Find the replicating portfolio: $x$ shares and a bond paying $y$ won at $T$. Describe it in words (buy or short how many shares; borrow or lend how much today).

(c) Price the call as the cost of that portfolio.

(d) Compute the risk-neutral probability $p$ and confirm the price with $e^{-rT}[pf_u + (1-p)f_d]$.

(e) A broker quotes the call at ₩ 3,500. Lay out the arbitrage: legs, cash today, and the payoff in each state at $T$.

(f) In one sentence: why did the probability of an up move never enter (a)–(e)?

## Problem 2 — The same call the hard way (DCF)

Investors require $\alpha = 12\%$ per annum on the stock in Problem 1.

(a) Find the real-world probability $p^*$ of an up move. Compare it with $p$.

(b) Using the replicating portfolio from Problem 1, compute the weights on stock and bond and the call's required return $r_{\text{call}}$.

(c) Price the call by DCF with $p^*$ and $r_{\text{call}}$. Does it match Problem 1?

(d) The stock rises to ₩ 62,000 one minute later, everything else unchanged. Without computing, say whether $r_{\text{call}}$ rises or falls, and why DCF is therefore impractical for options.

## Problem 3 — Any payoff will do: the ELS cliff

Last week's ELS (Problem 3 of PS 8) was a bond, a short put, and a short *cash-or-nothing* put: the saver forfeits a fixed amount — the 59 cliff — if the index ends below the barrier. Seen from the holder's side, a cash-or-nothing put pays a fixed amount if the index ends below a barrier and nothing otherwise. On the tree of Problem 1, consider a security that pays ₩ 10,000 if $S_T < 63{,}000$ and nothing otherwise.

(a) Find the replicating portfolio ($x$, $y$). Which way is the share position, and why?

(b) Price the security two ways: as the cost of the portfolio and with $p$.

(c) In one sentence: what does this problem say about which payoffs a binomial tree can price?

## Problem 4 — Two steps: call, put, parity, and early exercise

A NYSE-listed Korean e-commerce stock trades at \$20. Over each of the next two 6-month periods it either rises 25% or falls 20%; $r = 4\%$. Consider 1-year European options with $K = 20$.

(a) Draw the tree and compute the risk-neutral probability $p$.

(b) Price the call by backward induction, showing the value at each node.

(c) Price the put the same way.

(d) Verify put-call parity with your two prices.

(e) Now suppose the put is American. At which node, if any, is it exercised early, and what is it worth?

(f) Compute the call's hedge ratio $x$ at each of the three non-terminal nodes. What happens to it after an up move, and after a down move?

## Problem 5 — The ELS desk

Ms. Park runs the equity-derivatives desk of a Yeouido securities firm. Her firm has issued ELS whose embedded put is, in net, a European put on 100 million units of the KOSPI 200 (each unit pays $\max(380 - S_T, 0)$ won). The index is at 400 and moves up or down 15% every six months; $r = 3\%$; the put has one year to run. The desk *self-hedges* with KOSPI 200 futures, which have a multiplier of ₩ 250,000 per index point.

(a) Draw the tree and price the put per unit of index.

(b) Compute the put's hedge ratio $x$ at today's node and at each 6-month node.

(c) The desk is *long* the puts (its customers sold them). How many index units of futures must it be long today to be hedged? How many contracts is that?

(d) Six months later the index is at 340. How many contracts must the desk now hold, and how many does it buy?

(e) Approximately how much variation margin did the desk pay on its original futures position as the index fell from 400 to 340? Explain in one sentence why (d) and (e) land on the same day, and why that was the problem in March 2020.

(f) Some smaller firms instead buy the identical put from a foreign investment bank ("back-to-back" hedging). In one sentence, what have they given up and what have they gained?

## Problem 6 — Four steps: count the paths

A stock trades at \$50. Each quarter it rises 10% or falls 10%; $r = 4\%$. A 1-year European call has $K = 50$.

(a) Compute $p$ for one quarterly step.

(b) Tabulate, for each end node $j = 0, \ldots, 4$: the number of paths, the risk-neutral probability, $S_T$, and the payoff.

(c) Price the call with the $N$-step formula.

(d) A classmate forgets the path counts and uses $p^j(1-p)^{N-j}$ as the node probabilities. What price does she get, and what is the quickest way to see her probabilities are wrong?

(e) What is the risk-neutral probability that the call finishes in the money? Is this the real-world probability? One sentence.

## Problem 7 — Index put with a dividend yield

The KOSPI 200 is at 400 with volatility 20%; $r = 3\%$ and the index's dividend yield is $q = 2\%$. Use 6-month steps.

(a) Compute $u$, $d$ and $p$.

(b) Price a 1-year *European* put with $K = 400$.

(c) Price the same put if it is *American*. Where is it exercised?

(d) A trainee forgets the dividend yield and uses $p = (e^{r\Delta t} - d)/(u - d)$. Is the put over- or under-priced? One sentence of intuition.

## Problem 8 — Currency and futures: change what the asset grows at

(a) USD/KRW is 1,400 with volatility 10%. The won rate is 3% and the dollar rate is 4.5%. For a tree with monthly steps, compute $u$, $d$ and $p$. Why is $p$ below one half, and what does that say about the risk-neutral drift of the won price of a dollar?

(b) KOSPI 200 futures trade at 400 with volatility 20%; $r = 3\%$. Using two 3-month steps, price a 6-month European call on the futures with $K = 400$.

(c) In (b), $p$ did not depend on $r$ at all. In one sentence, why not — and where does $r$ still enter?

## Problem 9 — Equity is a call on the firm

Hanbit Shipping has assets worth ₩ 80bn and a single zero-coupon bond with face value ₩ 70bn due in two years. Asset volatility is 25%; $r = 4\%$. Use annual steps with $u = e^{\sigma\sqrt{\Delta t}}$.

(a) Draw the asset tree and write the shareholders' payoff at each end node.

(b) Value the equity.

(c) Value the debt two ways: from the balance-sheet identity, and by discounting the creditors' payoff $\min(V_T, L)$ on the tree.

(d) What yield does the debt carry, and what is the credit spread over $r$?

(e) The CEO proposes selling the tankers and buying a fleet of LNG carriers, which leaves asset value unchanged but raises asset volatility to 45%. Recompute (b)–(d). Who gains, who loses, and by how much? What clause would the bondholders have wanted in the indenture?

## Problem 10 — True or false

Say whether each statement is true or false, with one sentence of reasoning.

(i) If investors become more optimistic about a stock, so that the real probability of an up move rises with $S_0$, $u$, $d$ and $r$ unchanged, the binomial call price rises.

(ii) If the stock's required return exceeds the risk-free rate, the risk-neutral probability of an up move is below the real one.

(iii) In a binomial tree, an American call on a non-dividend stock is never exercised early.

(iv) Doubling the number of steps, with $\sigma$ and $T$ fixed, halves the size of each up move $u - 1$.

(v) Risk-neutral valuation assumes that investors are indifferent to risk.

(vi) The replicating portfolio of a long put is a short position in the stock plus lending.

(vii) As the number of steps goes to infinity, the CRR tree price converges to the Black–Scholes price.